Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/31/2006
PI Name:	Packard, Richard E. Ph.D.		
Project Title:	Superfluid gyroscopes in space		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Berkeley	**State:**	CA
Zip Code:	94720	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-03
Start Date:	02/01/1998	**End Date:**	01/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and uncover the mysteries of the universe obscured by Earth's gravity		
Task Description:	1. The pick-up loop of a 4He phase-slip gyroscope will be increased until environmental noise or intrinsic fluctuations limit the rotational sensitivity. 2. A superfluid analog of a dc-SQUID will be built. The device will incorporate two Josephson weak links and will detect the Earth's rotation. 1. To extend the sensitivity of the 4He phase-slip gyroscope to its intrinsic limit. 2. To make a proof-of-principle demonstration of a 3He quantum interference gyroscope. 3. To develop sensitive techniques to monitor the critical current in a superfluid dc SQUID 4. To determine the noise limitations of a superfluid gyroscope 5. To design a scaled-up instrument useful for geodesy 6. To observe and study Josephson phenomena in He4 weak link arrays to assess the suitabilty of a He4 dc squid gyroscope. 7. To develop a dc chemical potential drive in He4 A superfluid gyroscope that can detect 10-9 of the Earth's rotation would provide an important complement to existing		

	geodesy technology based on very long- baseline interferometry. Since the Earth's rotation vector changes both in magnitude and direction, these quantities must be monitored for input data into GPS and other navigational systems. These changes also reflect processes both on the surface and interior of the Earth. the superfluid gyroscope might also have impact in seismology on Earth or on other planets.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Precision measurements of the Earth's rotation. Observation of the rotational component of seismic waves reflects the structure of the Earth's mantle.
Task Progress:	We have designed and constructed a cell to determine the current-phase relation for He3 weak links at elevated pressure. We have developed a weak link array apparatus that exhibits Josephson oscillations in He4. This is the first observation of this important phenomenon.
Bibliography Type:	Description: (Last Updated: 09/12/2008)
Articles in Peer-reviewed Journals	Hoskinson E, Packard R, Haard TM. "Oscillatory motion: Quantum whistling in superfluid He4." Nature 2005 Jan 27;433(7024):376. PMID: 15674281 http://dx.doi.org/10.1038/433376a , Jan-2005
Articles in Peer-reviewed Journals	Penanen K, Hoffmann JA, Davis JC, Packard RE. "Effect of 3He on third sound attenuation in thick 4He Films." Journal of Low Temperature Physics 2004 Mar;134(5-6):1069-77. http://dx.doi.org/10.1023/B:JOLT.0000016730.44519.37 , Mar-2004
Articles in Peer-reviewed Journals	Hoffmann JA, Penanen K, Davis JC, Packard RE. "Measurements of attenuation of third sound: Evidence of trapped vorticity in thick films of superfluid 4He." Journal of Low Temperature Physics 2004 May;135(3-4):177-202. http://dx.doi.org/10.1023/B:JOLT.0000024548.03525.cd , May-2004
Articles in Peer-reviewed Journals	Packard RE. "Phase slips and Josephson weak links in superfluid helium." Journal of Low Temperature Physics 2004 Jun;135(5-6):471-8. http://dx.doi.org/10.1023/B:JOLT.0000029508.81716.5c , Jun-2004
Articles in Peer-reviewed Journals	Ishiguro R, Ishikawa O, Yamashita M, Sasaki Y, Fukuda K, Kubota M, Ishimoto H, Packard RE, Takagi T, Ohmi T, Mizusaki T. "Vortex formation and annihilation in three textures of rotating superfluid 3He-A." Phys Rev Lett. 2004 Sep 17;93(12):125301. PMID: 15447273 , Sep-2004